



Exploring technology acceptance patterns of users of the mixed-reality sport technology Zwift: antecedents and consequences of technology acceptance

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Abstract

The use of digital technologies in sports is increasing. The current paper deals with the technology acceptance of users of a mixed-reality sport technology called Zwift. The purpose of this study is to identify distinct profiles of Zwift users based on indicators of technology acceptance. Latent profile analysis ($n = 877$) identified three distinct latent profiles representing subgroups characterized by moderate, high and very high acceptance indicators. Results of the multinomial regression show that individuals with higher perceptions of the basic psychological need for autonomy and relatedness as well as individuals with high data privacy perceptions are more likely to be classified in the very high and high acceptance profiles than in the moderate acceptance profile. The results also show that older individuals are more likely to be classified in the moderate acceptance profile than in the very high acceptance profile even though odds ratio is rather low. The high and very high acceptance profiles also show higher behavioural intention to use Zwift in the future than the moderate acceptance profile.

Keywords Technology acceptance · Mixed-reality sport technology · Zwift · Latent profile analysis

1 Introduction

In recent years, the use of digital technologies in different areas of sports has been increasing [1]. Digital technologies have led, for example, to the formation of e-sports [2], the development of wearable fitness devices which are used to track and monitor users' physical activity and training-related data [3], virtual- or mixed-reality environments for improving sports skills [4, 5] or the launch of virtual-/mixed-reality sports competitions [6].

Different realities could be classified based on the reality–virtuality continuum by Milgram, Kishino [7]. Real environments with real objects and virtual realities with computer-generated objects are seen as the endpoints of the continuum. Mixed realities are located between those poles [7, 8]. In virtual realities (VR), individuals interact and navigate in real time in computer-generated environments [4, 9], “providing a sensory immersive experience”

[8], whereas in mixed realities (MR) “real and virtual objects were merged” [8].

Current research in the context of virtual or mixed realities in sports focuses on motivational issues of technology usage [10], cognitive functions (e.g. the difference between virtual and real sport environments) [11, 12], motor skills learning and sports performance [13–16], and emotional issues [17, 18] in diverse populations.

Through the rise of digital technologies in different areas of our lives (especially during the COVID-19 pandemic), a wide range of research is dealing with the acceptance and usage of new technologies in different contexts like health [19, 20], fitness apps and wearables [21–23], learning management systems [24, 25] or virtual environments [26–29]. Technology acceptance factors shape individuals' decision-making, behavioural intention and usage of a technology [30].

Although there is a huge body of research dealing with technology use and acceptance of diverse technologies [30], few studies have addressed technology acceptance issues of virtual- or mixed-reality sport applications [28, 29].

The current paper refers to a mixed-reality (MR) sport environment called Zwift [31]. In this interactive online cycling platform, users' real-world cycling performances (on

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a smart trainer) are transferred into a virtual environment where different users, who have the same hardware (interactive smart trainer which is connected to their own bike), interact [31, 32]. The user can customize their avatars, join group riding events and competitions or use it for personal trainings sessions. [For a detailed description of Zwift, see 31.] Even if technologies are effective and well constructed, they are not always accepted by users or potential users [28]. In the growing body of the literature dealing with VR/MR in sports [4, 6, 10], there is less research on technology acceptance [e.g. 28]. The aim of the present study is to examine whether Zwift users could be classified into distinct subgroups based on indicators of technology acceptance. Furthermore, antecedents (motivational constructs, age and data privacy perceptions) and consequences (intention to use the technology in future) of these technology acceptance patterns are surveyed.

2 Technology acceptance indicators and their associations with behavioural intention

The Technology Acceptance Model [33, 34], the Unified Theory of Acceptance and Usage of Technology (UTAUT) [35] and extended versions of the frameworks [30, 36] are prominent and widely used frameworks in studying technology acceptance and use issues [see 30, 37]. These frameworks describe psychological determinants which have an impact on users' acceptance and behavioural intention to use (and the actual use) a technology. According to the UTAUT framework and its extensions [30, 36], performance expectancy, effort expectancy and social influence are related to behavioural intentions to use a technology and this intention, as well as facilitating conditions (users' perceptions of the availability of resources and support), is associated with technology use. Performance expectancy is the individual's belief that the technology will be useful in performing a certain activity [36] and is the most important predictor of the intention to use the technology [35]. Effort expectancy is defined as the ease of use [36]. Social influence focuses on the social environment of the user. It is the user's perception that friends, etc. believe that they should use the technology [36].

Empirical work in the context of VR or MR technologies shows positive connections between the intention to use the technology and the constructs of performance expectancy/perceived usefulness [26–28, 38–40] and ease of use [26–28]. Findings from studies in the fitness app context also show the connections between behavioural intention and the three variables performance expectancy, effort expectancy and social influence [41, 42].

In the extended UTAUT 2 framework [36], the effects of hedonic motivation (perceived enjoyment when using a technology), price value, facilitating conditions and habit on behavioural intention were included into the framework. Empirical work shows that perceived enjoyment is a significant predictor of the intention to use VR in sports [28] and health contexts [43]. Furthermore, the social presence of others in virtual environments and social interactions increases users' enjoyment in VR environments, which is positively connected to behavioural intentions [44]. Social factors as well as performance improvement, health-related motives (increase one's physical activity and low risk of injuries) and customization are seen as important motives of Zwift users which are linked to continued use of this technology [10].

3 Factors associated with technology acceptance indicators

3.1 Users' basic psychological needs

Self-determination theory [45, 46] was used as the theoretical framework in studying the role of motivational variables in various technology contexts like e-learning [47–49], mobile-based assessment [50], information and communication technology [51], social exercise platforms [52] and wearable fitness devices [21]. Self-determination theory is a macrotheory of human motivation [53]. Referring to this theory, users are more likely to use a technology if the technology satisfies their basic psychological needs, namely autonomy, relatedness and perceived competence. The satisfaction of these basic psychological needs leads to self-determined forms of motivation [45, 53]. Zwift has the potential to satisfy users' needs for autonomy by providing freedom of choice concerning cycling activities (e.g. various virtual tracks; users can cycle when they want without any restrictions due to weather or daytime, etc.), relatedness through features like group ridings, being part of a social community and virtual challenges, and competency by defining goals and monitoring progress.

Research that applied self-determination theory in technology contexts shows that autonomy, perceived competence and relatedness are significant predictors of performance expectancy/perceived usefulness [51, 54] and perceived ease of use [50]. Autonomy and relatedness are positively linked to perceived enjoyment [51] and intrinsic motivation [55]. In the case of wearable fitness devices, the satisfaction of users' basic psychological needs for autonomy, relatedness and competence is positively linked to users' behavioural intention [21].

3.2 Privacy issues

Recent research into technology usage and acceptance also refers to the role of technology trust perceptions (e.g. perception of privacy/protection of one's own data) [21, 56]. For example, the perception of the privacy and security of one's data is assumed to build users' trust perception towards the technology [21, 57–59]. Privacy risks and loss of privacy when using a technology influences technology adoption and usage [58–60]. Empirical research in the health context shows that a user's perceived privacy concerns is negatively associated with the user's adoption of a wearable device [58, 60]. Privacy concerns do not always decrease the adoption of a technology. Kang, Jung [61] show that there is a subgroup of smartwatch users, which is characterized by high levels of privacy concerns and high perceptions of benefits when using the technology (ambivalent users). This group has shown high levels of technology (smart watch) usage. The authors called it the “smart wearables-privacy paradox” [61]. Based on these results, we incorporated privacy perceptions as an antecedent of technology acceptance and usage into our study. It is assumed that besides motivational constructs, privacy perceptions are associated with technology acceptance profiles.

3.3 Aim of the study

Recent technology acceptance studies primarily use variable-centred approaches such as structural equation modelling [e.g. 28, 38, 50] to examine associations between variables of interest. Person-centred approaches (grouping individuals qualitatively and quantitatively into subgroups regarding constructs/variables of interest) are less common in technology acceptance/usage studies [e.g. 61], but well established in social sciences [62–64]. These approaches are used to study individual differences by examining patterns of shared behaviour, etc. [65].

The present study seeks to address technology acceptance patterns of users of a mixed-reality cycling application called Zwift [31]. Therefore, a person-centred approach is used to discover latent subgroups concerning technology acceptance indicators in a sample of Zwift users and determine whether these subgroups differ in their behavioural intention to use the technology in future. To get information on the antecedents of subgroup membership, motivational constructs based on self-determination theory [45], age and data privacy perceptions are surveyed.

The following research questions were examined:

Research question 1: Which distinct latent profiles of Zwift users regarding their technology acceptance indicators (performance expectancy, effort expectancy, social influence, perceived enjoyment) can be identified using a person-centred approach?

Research question 2: Are there significant differences between the latent profiles in the intention to use the technology in future?

Research question 3: Can profile membership be predicted by users' perception of basic psychological needs satisfaction (autonomy, relatedness and competence), age and privacy perceptions?

4 Materials and method

4.1 Sample and data collection

An online questionnaire was used for data collection. The link to the online questionnaire was spread in sports-specific social media groups and personal networks during the COVID-19 pandemic. Sports clubs and federations (bicycle, triathlon) were contacted and asked whether they would send out the link to their members. The clubs and federations, as well as a sports magazine with high range, posted the link on their social media channels; 877 currently active and former active Zwift users ($M_{\text{age}} = 39.1$ years; $SD_{\text{age}} = 9.8$ years) from Austria and Germany participated in the study: 22.7% of the participants were female ($M_{\text{age}} = 38.3$ years; $SD_{\text{age}} = 9$ years) and 77.3% were male ($M_{\text{age}} = 39.4$ years; $SD_{\text{age}} = 10$ years). There are no significant age ($U = 17,682$, $p = 0.694$; active users: $M = 39.2$ years; $SD = 9.8$ years; former users: active users: $M = 38.4$ years; $SD = 9.9$ years) or gender differences (Fisher's exact test: $p = 0.356$) between currently active and former active Zwift users. The characteristics of the sample are displayed in Table 1, showing that 89.5% have used Zwift more than 20 times. On average, 66% ($SD = 28.8$) of participants cycling training per week was completed using Zwift.

4.2 Measures

The wording of the items used in this study was adjusted to the Zwift technology. Intercorrelations and internal consistency of the scales are shown in Table 2.

Performance expectancy: Performance expectancy was measured using 4 items (example item: “Using Zwift increases the effectivity of my training”; 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree) adopted from Lee et al. [51] and Venkatesh et al. [35].

Effort expectancy: Effort expectancy was measured using a 5 items scale from Lee et al. [51] and Venkatesh et al. [35] (example item: “I find that Zwift is easy to use”; 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree).

Social influence: The social influence scale consists of 4 items (example item: “People who influence my behaviour think that I should use Zwift”, 5-point rating scale

Table 1 Sample characteristics

Characteristics	n (%)
<i>Gender</i>	
Female	199 (22.7%)
Male	678 (77.3%)
<i>Age groups</i>	
18–20 years	11 (1.3%)
21–30 years	185 (21.1%)
31–40 years	290 (33.1%)
41–50 years	264 (30.1%)
51–60 years	114 (13%)
61–70 years	13 (1.5%)
<i>Experience with Zwift (How often the users have used Zwift)</i>	
1 to 5 times	24 (2.7%)
6 to 20 times	67 (7.6%)
More than 20 times	785 (89.5%)
Missing values	1 (0.1%)
Number of currently active Zwift Users	833 (95%)
Number of former Zwift Users (currently not using Zwift)	44 (5%)
<i>Training sessions per week</i>	
Daily	226 (25.8%)
5 to 6 times per week	434 (49.5%)
3 to 4 times per week	200 (22.8%)
1 to 2 times per week	16 (1.8%)
less than once a week	0 (0%)

n = 877

ranging from 1 = totally disagree to 5 = totally agree) and was adopted from Venkatesh et al. [35].

Perceived enjoyment: Perceived enjoyment was measured using the 4 item scale (example item: “Using Zwift is enjoyable”, 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree) described in Lee et al. [51] and Venkatesh et al. [35].

Basic psychological needs (all scales use a 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree): To measure the perceived satisfaction of participants basic psychological needs for *autonomy* (4 items; example item: “Using Zwift, I can make meaningful choices about my sport activity”), *relatedness* (4 items; example item: “Using Zwift, would help me exercise with others”) and *competency* (4 items; example item: “Using Zwift, I can be successful creating physical activity and sport related goals”) an adapted version of the Wearable Technology Motivation Scale was used [21].

Privacy (security of personal data): The privacy scale (3 items; example item: “I feel that Zwift will keep my data secure”; 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree) was adopted from the Wearable Technology Trust Scale [21].

Behavioural intention: A single item (“I plan to use Zwift in the next 3 months”; 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree) was used to measure the behavioural intention [adopted from 35].

4.3 Analytic approach

First, confirmatory factor analysis was conducted using diagonally weighted least squares estimator (DWLS). Therefore, the *R* [66] package *lavaan* [67] was used. Goodness of fit of the measurement model was assessed using the root mean square error of approximation (RMSEA), standardized root mean square residual (SRMR), Tucker–Lewis index (TLI) and comparative fit index (CFI). The following cut-off criteria provided by Hu, Bentler [68] were used: $SRMR \leq 0.08$, $RMSEA \leq 0.06$; CFI and TLI ≥ 0.95 indicate a good model fit.

The aim of this paper is to identify unobserved subgroups of Zwift users based on their technology acceptance patterns. Therefore, latent profile analysis (LPA), a latent variable modelling technique [64, 69], was used. The assumption of

Table 2 Cronbach’s alpha and intercorrelations among the variables

	Cronbach’s alpha	1	2	3	4	5	6	7
1. Performance expectancy	.891	–	.342**	.451**	.663**	.367**	.455**	.222**
2. Effort expectancy ^b	.813		–	.133**	.351**	.084*	.162**	.249**
3. Social influence	.754			–	.362**	.563**	.297**	.227**
4. Perceived enjoyment	.91				–	.473**	.426**	.313**
5. Relatedness	.786					–	.346**	.204**
6. Autonomy	.878						–	.183**
7. Privacy	.705							–

n = 877; Standardized factor scores from CFA were used

Effort expectancy: ^bhigh values on the effort expectancy scale displays low effort expectancy perceptions; competency scale was excluded due to low internal consistency of .362

p* < .05; *p* < .01

Table 3 Fit indices of the LPA models with increasing numbers of latent profiles

No. latent profiles	Log-likelihood	BIC	Entropy	Lo–Mendell–Rubin adjusted likelihood ratio test (<i>p</i>)	Smallest class size % (<i>n</i>)
2	–3431.779	6978.758	.803	0.015	285 (32.5%)
3	–3294.492	6765.174	.753	0.000	175 (19.9%)
4	–3216.204	6669.587	.802	0.062	6.1 (54%)

The three-profile solution shows the best model fit

BIC Bayesian information criterion; *n* = 877

LPA is that individuals can be classified in subgroups (called latent profiles) which are characterized by different personal attributes [63]. “LPA is particularly useful for researchers in social sciences as patterns of shared behaviour between and within samples may be missed when researchers conduct interindividual, variable-centred analyses” [65]. Models with 2-*x* latent profiles were performed using *Mplus* 8 [70]. To avoid local maxima solutions, multiple random starting sets were used (starts = 5000 500; iterations = 200) [71]. The best fitting model was chosen by inspecting the Bayesian information criterion (BIC) [72], entropy indicator [73] and the Lo–Mendell–Rubin adjusted likelihood ratio test [74].

Smaller values in the Bayesian information criterion (BIC) indicate better model fit [71] and higher entropy values (ranging from 0 to 1) indicate better classification accuracy. The Lo–Mendell–Rubin adjusted likelihood ratio test of model fit (LMR test) denotes whether there is a significant improvement of model fit by adding a further latent profile. A significant LMR test shows that the *k*-profile LPA solution has a better model fit than the *k*–1 profile solution [75].

Differences between the latent profiles in individuals’ behavioural intention (research question 2) to use Zwift in the next 3 months were assessed using Kruskal–Wallis tests.¹ To identify significant predictors of latent profile membership (research question 3), multinomial regression analysis was performed using *Mplus* (using the 3 Step approach; see [76]).

5 Results

5.1 Preliminary analyses

First, the reliability and validity of the measurements were assessed. The factor structure of all measures was analysed using confirmatory factor analysis (CFA). One item (competency scale) had to be excluded because of its low factor loading of 0.31. The CFA shows a good

model fit ($\chi^2(375) = 665.195$, $p = 0.000$; RMSEA = 0.03, SRMR = 0.042, CFI = 0.997, TLI = 0.997). Standardized factor loadings are between 0.451 and 0.933. The reliabilities of all scales, with the exception of the competency scale, are good (performance expectancy: $\alpha = 0.891$; effort expectancy $\alpha = 0.813$; social influence: $\alpha = 0.754$; perceived enjoyment: 0.91; relatedness: $\alpha = 0.786$; autonomy: $\alpha = 0.878$; privacy: $\alpha = 0.705$; see Table 2). The internal consistency of the competency scale ($\alpha = 0.362$) is very low. Competency scale was excluded from further analyses. Standardized factor scores derived from the CFA were used for all analyses.

5.2 Latent profile analyses

Research Question 1: Which distinct latent profile of Zwift users regarding their technology acceptance indicators (performance expectancy, effort expectancy, social influence, perceived enjoyment) can be identified using a person-centred approach?

Three distinct latent profiles were identified. The Lo–Mendell–Rubin adjusted likelihood ratio tests indicate that the three-profile solution shows the best model fit (see Table 3). Adding a fourth profile would not significantly improve the model fit. The three-profile solution shows acceptable classification accuracy (entropy = 0.753) as well as acceptable size of the smallest profile. Analysis of variance and Kruskal–Wallis tests show that all latent profiles significantly differ in the variables used for LPA (see Table 4). The three latent profiles presented in Figs. 1 and 2 are described in the following.

Estimated means of the standardized factor scores derived from CFA are displayed in Fig. 1. Furthermore, unstandardized mean composite scores (see Fig. 2) are calculated to obtain information about the acceptance indicators in each profile with values ranging from 1 to 5 (5-point rating scale was used; 1 = totally disagree to 5 = totally agree). Both values are used to describe the latent profiles.

¹ ANOVAs could not be performed due to unequal variances.

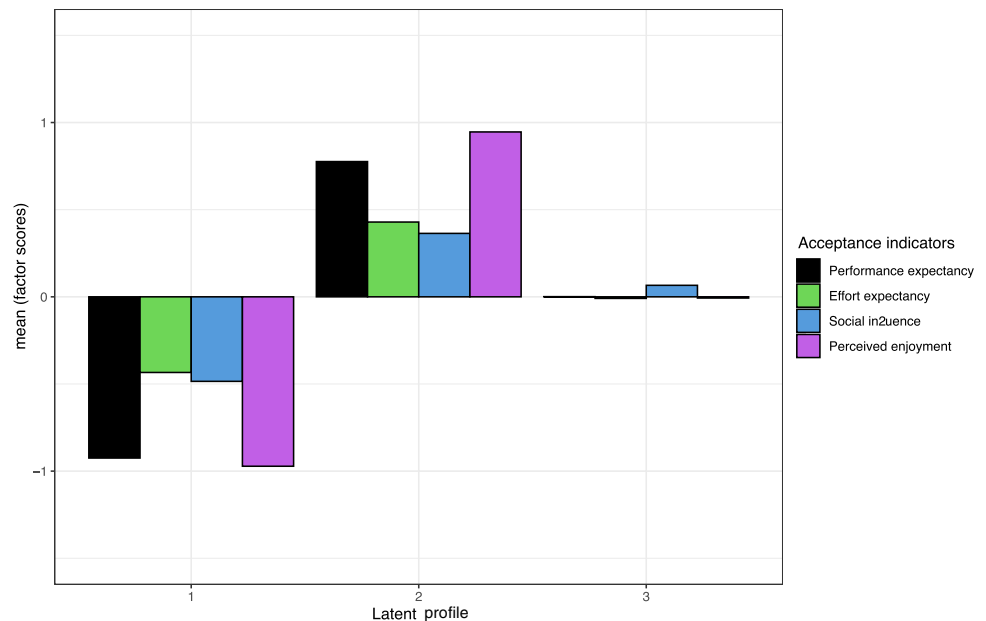
Table 4 Acceptance indicator; differences between the latent profiles

LPA Indicators	Profile 1 (factor scores; $n=206$)		Profile 2 (factor scores; $n=175$)		Profile 3 (factor scores; $n=496$)		ANOVA/Kruskal–Wallis test ¹	
	M	SD	M	SD	M	SD	$F(df)/H(df)^a$	Significant pairwise comparisons
Performance expectancy	−0.925	0.528	0.776	0.216	0.001	0.424	597.401(2)*** ^a	1 < 2; 1 < 3; 3 < 2
Effort expectancy ^b	−0.434	0.69	0.429	0.627	−0.009	0.655	81.789(2)***	1 < 2; 1 < 3; 3 < 2
Social influence	−0.485	0.511	0.364	0.678	0.066	0.52	185.402(2)*** ^a	1 < 2; 1 < 3; 3 < 2
Perceived enjoyment	−0.972	0.604	0.946	0.21	−0.007	0.507	573.836(2)*** ^a	1 < 2; 1 < 3; 3 < 2

$n=877$; M =estimated mean; SD =estimated standard deviation; factor scores from CFA were used; ^bhigh values on the effort expectancy scale display low effort expectancy perceptions

* $p < .05$; ** $p < .01$; *** $p < .001$; pairwise comparisons: Bonferroni or Mann–Whitney- U -tests

^aKruskal–Wallis test $H(df)$; Kruskal–Wallis test were used because of unequal variances; Profile 1: moderate acceptance profile; Profile 2: very high acceptance profile; Profile 3: high acceptance profile

Fig. 1 Estimated means of the three-profile solution, standardized factor scores (color figure online)

5.2.1 Latent Profile 1: moderate acceptance profile

Individuals classified in latent Profile 1 show the lowest values in all scales. Mean composite scores (ranging from 1 to 5; lower values indicate low performance expectancy, high effort expectancy, low social influence and low perceived enjoyment) show that they have relatively moderate to relatively high performance expectancy ($M=3.7$; $SD=0.73$) and they perceive that using Zwift takes little to moderate effort ($M=3.77$; $SD=0.74$). Furthermore, individuals in Profile 1 show moderate values in the perceived enjoyment scale ($M=3.39$; $SD=0.72$) and relatively low social influence values ($M=2.33$; $SD=0.8$).

5.2.2 Latent Profile 2: very high acceptance profile ($n=175$)

Individuals showing the highest values in all scales are grouped together in Profile 2. This group shows very high performance expectancy ($M=4.99$, $SD=0.03$), very low effort expectancy ² ($M=4.49$, $SD=0.55$) and very high perceived enjoyment values ($M=4.98$, $SD=0.07$). Individuals report the highest social influence values even though mean composite scores are average ($M=3.3$, $SD=1.01$).

² high values represent low effort expectancy perceptions

Fig. 2 Estimated means of the three-profile solution, composite scores (5-point rating scale was used; 1 = totally disagree to 5 = totally agree; high values on performance expectancy, social influence and perceived enjoyment represents high performance expectancy, social influence and perceived enjoyment perceptions; high values on the effort expectancy scale represent low effort expectancy perception) (color figure online)

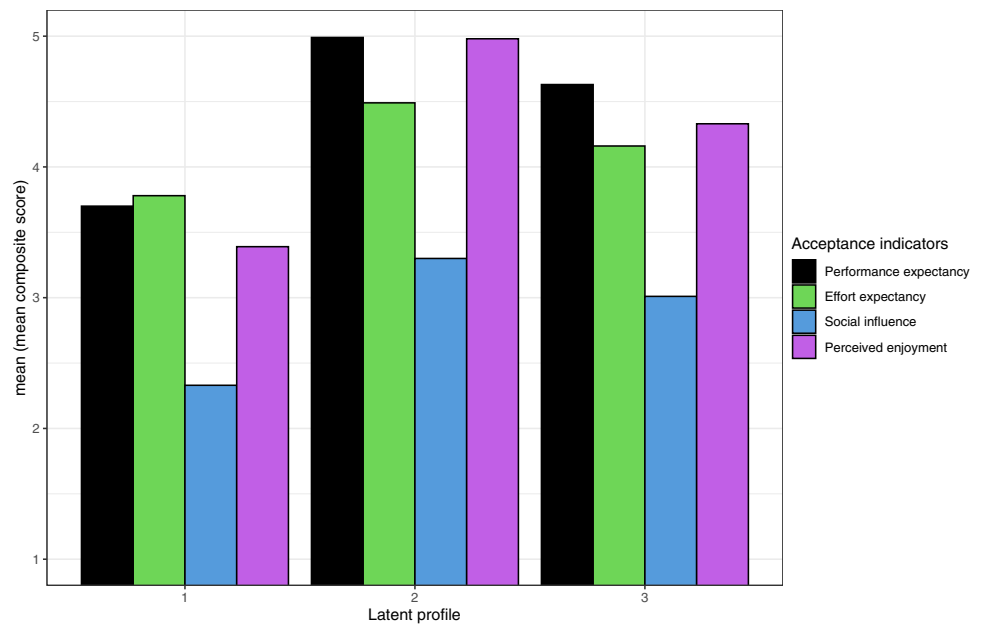


Table 5 Multinomial regression

		Estimate	SE	Estimate/SE	Two-tailed <i>p</i> value	OR
Profile 1 versus Profile 2 ^a	Age	−0.026	0.017	−1.493	0.135	0.975
	Basic psychological need: relatedness	1.843	0.268	6.866	0.000	6.315
	Basic psychological need: autonomy	2.026	0.332	6.101	0.000	7.585
	Technology trust: privacy	1.198	0.260	4.614	0.000	3.312
Profile 1 versus Profile 3 ^a	Age	−0.041	0.014	−2.933	0.003	0.96
	Basic psychological need: relatedness	1.070	0.184	5.819	0.000	2.915
	Basic psychological need: autonomy	0.841	0.175	4.795	0.000	2.318
	Technology trust: privacy	0.933	0.209	4.747	0.000	2.699
Profile 2 versus Profile 3 ^b	Age	−0.016	0.012	−1.308	0.191	0.984
	Basic psychological need: relatedness	−0.773	0.197	−3.929	0.000	0.462
	Basic psychological need: autonomy	−1.185	0.279	−4.242	0.000	0.306
	Technology trust: privacy	−0.205	0.171	−1.198	0.231	0.815

^a = Parameterization using reference profile 1

^b = parameterization using reference profile 2; *Estimate* = regression coefficient *B*

OR odds ratio, *SE* standard error

Profile 1: moderate acceptance profile; Profile 2: very high acceptance profile; Profile 3: high acceptance profile; *n* = 877

5.2.3 Latent Profile 3: high acceptance profile (*n* = 496)

Individuals classified in Profile 3 are characterized by high performance expectancy ($M = 4.63$, $SD = 0.39$), perceived enjoyment ($M = 4.33$; $SD = 0.49$) and low effort expectancy ($M = 4.16$, $SD = 0.64$). These values are not as high as the values of individuals in Profile 2. Furthermore, they report moderate social influence values ($M = 3.01$, $SD = 0.8$).

5.3 Predictors of profile membership

Multinomial regression analysis was performed to identify significant predictors of profile membership (see Table 5). The results show that older individuals are more likely to be classified to Profile 1 than to Profile 3 (estimate = -0.041 , $p = 0.003$, $OR = 0.96$) even though odds ratio is rather low. Individuals who report that using the technology satisfies

Table 6 Behavioural intention to use Zwift in the future; differences between the latent profiles

Variable	Profile 1 (<i>n</i> = 206)		Profile 2 (<i>n</i> = 175)		Profile 3 (<i>n</i> = 496)		Kruskal–Wallis test	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>H</i> (<i>df</i>)	Significant pairwise comparisons
Behavioural intention ^a	4.11	1.19	4.89	0.41	4.77	0.63	124.634(2) ^{***}	1 < 2; 1 < 3

M = estimated mean; *SD* = estimated standard deviation

^abehavioural intention item: “I plan to use Zwift in the next 3 months”; 5-point rating scale ranging from 1 = totally disagree to 5 = totally agree

p* < .05; *p* < .01; ****p* < .001; Levene test shows heterogeneity of variances so Kruskal–Wallis test was performed; pairwise comparisons: Mann–Whitney-*U*-test (with Bonferroni correction)

Profile 1: moderate acceptance profile; Profile 2: very high acceptance profile; Profile 3: high acceptance profile; *n* = 877

their basic psychological needs of autonomy and relatedness to a higher degree are more likely to be classified in the high acceptance Profile 3 (relatedness: estimate = 1.07, *p* = 0.000, *OR* = 2.915; autonomy: estimate = 0.841, *p* = 0.000, *OR* = 2.318) and the “highest values in all scores” Profile 2 (relatedness: estimate = 1.843, *p* = 0.000, *OR* = 6.315; autonomy: estimate = 2.026, *p* = 0.000, *OR* = 7.585) than in Profile 1 representing the lowest acceptance indicator scores in the LPA indicators. Furthermore, individuals scoring high on the basic psychological needs of autonomy and relatedness are more likely to be classified in Profile 2 than in Profile 3 (relatedness: estimate = −0.773, *p* = 0.000, *OR* = 0.462; autonomy: estimate = −1.185, *p* = 0.000, *OR* = 0.306). The odds ratios are rather low.

Individuals reporting higher perceptions of data security are more likely to be classified in Profile 2 (estimate = 1.198, *p* = 0.000, *OR* = 3.312) or Profile 3 (estimate = 0.933, *p* = 0.000, *OR* = 2.699) than in Profile 1. The comparison of Profiles 2 and 3 with regard to data privacy is not significant.

5.4 Profile membership and behavioural intention (future)

Individuals were asked about their intention to use Zwift in future (see Table 6). The latent profiles significantly differ in their behavioural intention ($H(2) = 124.634$, *p* < 0.001). Post hoc analyses show that individuals of Profile 2 (*M* = 4.89, *SD* = 0.41, *U* = 10,578.5, *p* < 0.001) as well as Profile 3 (*M* = 4.77, *SD* = 0.63, *U* = 33,268.5, *p* < 0.001) report higher behavioural intention than individuals classified in Profile 1 (*M* = 4.11, *SD* = 1.19). Profile 2 and Profile 3 did not show significant differences in their behavioural intention.

6 Discussion

The present study seeks to identify distinct subgroups of Zwift users based on their technology acceptance indicators. The aim is to answer the question whether there are different

configurations of various technology acceptance indicators that characterize Zwift users, and whether motivational constructs, age and data privacy perceptions are associated with profile membership. Furthermore, differences in behaviour intention to use the technology in future between the subgroups are surveyed.

The results provide three key findings. First, Zwift users can be classified into three distinct subgroups regarding their technology acceptance. Individuals classified to latent Profile 1 show the lowest values in all indicators compared to the other profiles. The individuals in this group report that they have moderate to high perceptions concerning the performance of the technology, and that using the technology requires moderate to little effort. They also report moderate enjoyment when using the technology and that they perceive low social influence in using Zwift. In contrast, individuals who report very high performance expectancies and very low effort expectancies are grouped in Profile 2. These individuals also enjoy using this technology very much. The social influence values are higher than in Profile 1, but still on a moderate level. Individuals in Profile 3 are also characterized by high performance expectancies and low effort expectancies, but the values are lower than in Profile 2. They also enjoy using the technology and report that the social influence from others is moderate. So, this profile is similar to Profile 2, but with lower values in all indicators.

There was no profile representing low technology acceptance. Furthermore, the social influence scale has only moderate values even in the high acceptance profiles. It has to be mentioned that social media influences could also be relevant, which are not covered by the social influence scale used in this study. Further motives for using Zwift could be the possibility for customization [10] and the wide range of features of Zwift. The study of Wiesner et al. [77] shows that monitoring exercise levels, self-motivation, curiosity and personal health aspects are the main reasons of wearable fitness device users. These motives could also be important in acceptance and usage of the mixed-reality technology Zwift.

The existence of these subgroups leads to the question of antecedents and consequences of group membership. The second key finding is that even though there is no “low acceptance” subgroup, individuals’ perception of basic psychological needs satisfaction is associated with the acceptance patterns, and that these subgroups differ in their intention to use the technology in future. Current research shows that autonomy and relatedness are positively linked with technology acceptance indicators like performance expectancy, effort expectancy and perceived enjoyment [50, 51, 54]. The present study generally supports the role of the need for autonomy and relatedness in the technology acceptance context and expands our knowledge that autonomy and relatedness are not just linked with single acceptance indicators. Our results show the positive link of these basic psychological needs with more complex acceptance patterns derived from various acceptance indicators. Individuals with higher perceptions of autonomy and relatedness are more likely to be classified in the very high or high acceptance profiles than in the moderate acceptance profile. Furthermore, the very high acceptance Profile 3 also shows the highest intention to use the technology in future.

Especially when technologies take users seeking autonomy and relatedness into account when designing new technologies or features, the acceptance and behavioural intention to use the technology could be enhanced even in groups with relatively high usage behaviour. The results also show that older individuals are more likely to be classified to the moderate acceptance profile than to the very high acceptance profile, even though odds ratios are rather low.

As mentioned, we could not use the need for competency scale in our analyses due to the very low reliability of this scale. Referring to self-determination theory [45], all basic psychological needs should be satisfied to foster intrinsic motivation which should lead to high acceptance and behavioural intention. In particular, the need for perceived competency could be important in the context of Zwift because this mixed-reality has different features which help users to experience competency but also incompetency (individual training progresses, virtual races, etc.).

When studying technology acceptance and use, data privacy should be considered because low trust in data privacy issues could negatively influence individuals’ adoption and usage behaviour [58–60]. The third key finding of the present study is that individuals with high perceptions of data privacy are more likely to be classified in the high and very high acceptance profiles. The results are quite in line with current research [58, 60].

We conclude that the satisfaction of users’ needs for autonomy and relatedness in a mixed-reality sports seem to be more important for technology acceptance indicators than age and data privacy, even though they also shape users’

technology acceptance (see odds ratios in multinomial regression), and that individuals who are characterized by high acceptance indicators are more likely to use the system in future. The role of the need for competency remains unclear.

7 Limitations and future research directions

The present study is not free of limitations. First, we only asked the participants whether they plan to use Zwift in the future. To get information about users’ continuous use, longitudinal measures of usage behaviour would be more suitable due to the intention–behaviour gap [78]. Further research could also focus on the intention–behaviour gap and how VR or MR should be designed to make the translation of the intention into action more likely. Further research could focus on the experiences of individuals using VR or MR applications the first time (e.g. How do new users build technology trust? Which characteristics of users or applications foster or hinder trust formation? How do technology acceptance indicators of new users change over time?). The second limitation refers to the low reliability of the competency scale which had to be excluded from all analysis.

Author contributions MM, SM and FM wrote the main manuscript text. MM (corresponding author) performed the statistical calculations and wrote the major part the paper. SM collected the data and wrote parts of the paper. FM wrote parts of the paper. All authors made substantial contributions to the conception or design of the work and reviewed the manuscript.

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Data availability The data are not publicly available due to containing information that could compromise research participant privacy/consent.

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the University of Applied Sciences Wiener Neustadt, Austria.

Informed consent Informed consent was obtained from all individual participants included in the study.

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